Toward the Dynamical Convergence on the Jet Stream in Aquaplanet AGCMs

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ABSTRACT

Systematic sensitivity of the jet position and intensity to horizontal model resolution is identified in several aquaplanet AGCMs, with the coarser resolution producing a more equatorward eddy-driven jet and a stronger upper-tropospheric jet intensity. As the resolution of the models increases to 50 km or finer, the jet position and intensity show signs of convergence within each model group. The mechanism for this convergence behavior is investigated using a hybrid Eulerian–Lagrangian finite-amplitude wave activity budget developed for the upper-tropospheric absolute vorticity. The results suggest that the poleward shift of the eddy-driven jet with higher resolution can be attributed to the smaller effective diffusivity of the model in the midlatitudes that allows more wave activity to survive the dissipation and to reach the subtropical critical latitude for wave breaking. The enhanced subtropical wave breaking and associated irreversible vorticity mixing act to maintain a more poleward peak of the vorticity gradient, and thus a more poleward jet. Being overdissipative, the coarse-resolution AGCMs misrepresent the nuanced nonlinear aspect of the midlatitude eddy–mean flow interaction, giving rise to the equatorward bias of the eddy-driven jet. In accordance with the asymptotic behavior of effective diffusivity of Batchelor turbulence in the large Peclet number limit, the upper-tropospheric effective diffusivity of the aquaplanet AGCMs displays signs of convergence in the midlatitude toward a value of approximately $10^7\,\mathrm{m^2\,s^{-1}}$ for the ∇^2 diffusion. This provides a dynamical underpinning for the convergence of the jet stream observed in these AGCMs at high resolution.

1. Introduction

The jet stream is the most prominent feature in the midlatitude circulation and it plays a key role in orchestrating embedded synoptic weather systems and extreme meteorological events. As the grandest feature of the midlatitude atmospheric circulation, it has been simulated with reasonable realism since the earliest climate models (Manabe and Wetherald 1967). It might be a surprise to many that decades later, the most state-of-the-art global climate models, such as those participating in phase 5 of the the Coupled Model Intercomparison Project (CMIP5), are still suffering biases, often equatorward, in the location of the eddy-driven westerly jet (e.g., Kidston and Gerber 2010; Swart and Fyfe 2012). Because of the prominent influence of the jet stream on midlatitude climate features such as storm tracks and blocking, these biases have undermined the

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effectiveness of the climate models not only in simulating the present climate but also in projecting future trends, especially when the regional details are of concern.

Gerber et al. (2008) noticed that the bias in the latitude of the westerly jet can be manifested in the persistence of the phase of its leading mode of variability, which mainly reflects the sideways wobbling of the jet, with an equatorward biased jet corresponding to a longer persistence. In addition, evidence from phase 3 of CMIP (CMIP3) (Kidston and Gerber 2010) suggested that models with a more severe equatorward bias in the jet position tend to project a greater poleward jet shift under climate change driven by increasing greenhouse gas concentration. What determines the location of the jet stream, especially the eddy-driven component of it, is a complicated matter. Oftentimes, coarse horizontal resolution has been blamed for the jet biases. Comparing the higher- and lower-resolution simulations contributed from the same modeling group in the CMIP3 archive, Gerber et al. (2008) further noted that for the same model (in a sense that the dynamical core and physical parameterizations are the same) the coarser-resolution version simulates a more equatorward placed eddy-driven jet, a more persistent annular mode, and a larger annular mode trend under greenhouse gas forcing. Meanwhile, the jet biases in comprehensive climate models are more pronounced in the Southern Hemisphere, particularly in the summer when the midlatitude circulation is more zonally symmetric (e.g., Kidston and Gerber 2010; Arakelian and Codron 2012). This is consistent with idealized atmospheric general circulation model (AGCM) studies (e.g., Gerber and Vallis 2009) showing that the jet position and its variability are more sensitive in a zonally symmetric setting than those with large-scale zonal asymmetry in the forcing.

As such, this resolution dependency of the jet characteristics should be more readily identified in an aquaplanet configuration and this is the case indeed. Figure 1 shows the maximum speed of the 250-hPa-level jet versus the latitude of the peak of the near-surface (850 hPa) westerly wind, which indicates the location of the eddy-driven jet, simulated in three different aquaplanet AGCMs, each consisting of simulations at four different resolutions. In each of the model groups, there is clearly a propensity for the upper-level jet to weaken and the surface westerlies to shift poleward as the resolution doubles, with some exceptions between the penultimate and the higher resolutions. But the differences between the two highest resolutions are often statistically insignificant (see the crosses for the 99% confidence intervals for the jet position and intensity). By inspecting the meridional profiles of the near-surface

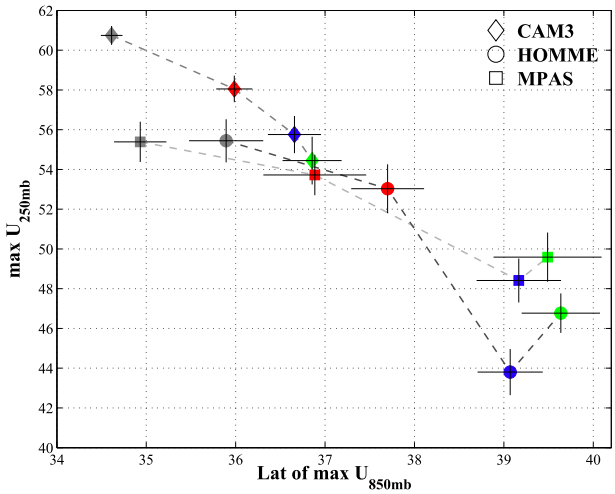

FIG. 1. Upper-level (250 hPa) jet intensity (m s^{-1}) vs the latitude of the near-surface (850 hPa) westerly wind maximum (°lat) in three AGCMs, each running at four different resolutions denoted by gray, red, blue, and green symbols in an order of increasing resolution. The crosses overlaid on each symbol gives the 99% confidence intervals based on Student's *t* test. The reduction of the effective sample size due to the autocorrelation of the jet indices is considered in the estimate of the confidence intervals according to Santer et al. (2000).

wind (an example is shown in Fig. 4a for the MPAS model), there appears a sign of convergence toward the "true" state of the jet of the respective model. In fact, similar convergence behavior was found in both full-physics AGCMs (Pope and Stratton 2002; Arakelian and Codron 2012) and those with a dry dynamical core (Wan et al. 2008) as the horizontal resolution of the AGCMs was systematically changed. Pope and Stratton (2002) attributed vaguely the sensitivity of the jet to the so-called nonlinear dynamics, while offering no specific explanation. Here we take as the first task of the current investigation to unveil this nonlinear dynamics that is responsible for the sensitivity of the jet stream to resolution. An interpretation will be provided in this study attributing the sensitivity shown in Fig. 1 to the dependence of effective diffusivity to model resolution.

A more fundamental challenge to the climate modeling community might be the following: As the climate models are run at increasingly finer resolution, can we determine whether and at what point they will converge on the key circulation features that matter to the mean climate and extremes, such as the intensity and the position of the westerly jet, atmospheric rivers, blockings, and spatiotemporal characteristics of the leading mode of the circulation variability, to name a few? We ask this question as we see some promising signs for the convergence of the jet stream in the aquaplanet AGCM simulations examined here, as well as evidence for the

improvement of the blocking activity (Matsueda et al. 2009; Jung et al. 2012), atmospheric rivers (Wick et al. 2013; Demory et al. 2014), and internal mode variability (e.g., Gerber et al. 2008; Arakelian and Codron 2012) as results of increased horizontal resolution.

Recent advances in geophysical fluid dynamics for two-dimensional turbulence offer a theoretical framework for quantitative assessment of the dynamical convergence on the processes (e.g., the jet stream) governed by "type II" flow, or the quasi-horizontal flow (e.g., Falkovich et al. 2001; Batchelor 1959; Haynes 2010). For this type of flow, the conservation of both energy and enstrophy demands a steep falloff in energy with decreasing scale (i.e., the characteristic -3 slope), so that the larger scales dominate the energy to a greater degree than in three-dimensional turbulence (Kraichnan 1967; Vallis 2006).

Moreover, for this type of flow there is no strong increase of velocity gradients as horizontal length scale shrinks; equivalently, the eddy turnover time and the stretching rate within this range are length scale invariant (Vallis 2006, their section 8). In contrast, the stretching rate for the "type I" flow, as exemplified by three-dimensional turbulent flow, scales as $\epsilon^{1/3}l^{-2/3}$, where ϵ is the energy input rate per unit volume at the injection scale. Therefore, the stretching rate increases as the eddy scale decreases (Haynes 2010). For the type-II flow, the stretching rate at a given scale is dominated by the "smooth" velocity gradient U/L (U and L represent the scales of the velocity and length, respectively, of the type-II flow) and the related effective diffusivity (denoted by K_{eff} and explained in section 3) is largely independent of the magnitude and nature of the ultimate dissipation. In particular, if the diffusion coefficient κ is small enough that the Peclet number (defined as $\mathrm{Pe} \equiv UL/\kappa$, a nondimensional parameter measuring the relative importance of the stirring to diffusion) is much larger than 1, the tracer field is predominantly controlled by the "chaotic advection" (Wiggins and Ottino 2004) by the smooth flow. The large Pe regime is often referred to as the "Batchelor turbulence" regime in the fluid dynamics community (e.g., Falkovich et al. 2001; Haynes 2010).

In numerical models, as the horizontal resolution keeps increasing, which allows smaller and smaller numerical diffusivity κ, the simulated circulation and tracer field approach the Batchelor regime and the effective diffusivity becomes independent of the exact value of κ. This asymptotic behavior of K_{eff} with increasing horizontal resolution has been demonstrated in both idealized chaotic advection calculation using a simple analytic flow (e.g., Shuckburgh and Haynes 2003) and more realistic calculation for the eddy-prevalent Southern Ocean (Marshall et al. 2006). What is remarkable about this property of K_{eff} is that the circulation features controlled by the quasi-two-dimensional eddies and the associated K_{eff}, such as the midlatitude eddy-driven jet, might converge as K_{eff} converges. On the other hand, for AGCMs with coarse and moderate resolutions, the numerical diffusion or the explicit hyperdiffusion used to prevent the accumulation of energy at the cutoff scales (Williamson 2008; Jablonowski and Williamson 2011) are often too diffusive (i.e., too large κ in the context of the discussion above); as a consequence, even the two-dimensional Batchelor regime is severely curtailed and the associated effective diffusivity is overestimated. The possible convergence on the effective diffusivity K_{eff} and the consequent impacts on the governing dynamics for the eddy-driven jet will be another focal theme of the current investigation.

We caution that the dynamical convergence in question here should not be confused with the numerical convergence discussed in previous studies (e.g., Boville 1991; Williamson 2008) or the issue of sufficient resolution of numerical schemes (e.g., Lander and Hoskins 1997). The entity of dynamical convergence in question (i.e., the mean westerly jet) is fully resolved numerically by the model; what is at issue is the faithful representation of the interscale interaction (involving energy and enstrophy cascades) that the dynamical convergence entails. Specifically, the question of dynamical convergence here is only pertinent in the sense that the scales of the phenomena in question are large enough that the dynamics of the two-dimensional Batchelor regime and its convergence properties[1] are relevant. For the case of the eddy-driven jet, its quantitative characteristics are set by the intricate interplay between the irreversible vorticity or quasigeostrophic potential vorticity (PV) mixing during wave breakings and the inhibition thereof due to the mixing barrier imposed by the jet itself. As to be shown in this paper, the value of the effective diffusivity serves as a useful metric for quantifying the convergence of the midlatitude jet dynamics.

The paper is arranged as follows. The models and the model datasets used in the current analysis are introduced in section 2. The theoretical framework (a hybrid Euler–Lagrangian finite-amplitude wave activity budget) based on the vorticity equation through a tracer-following coordinate transformation is developed in section 3, and some related mathematical derivations are archived in the appendixes. Section 4

[1] These properties shield the large-scale features from the uncertainties from the parameterized subgrid processes, thus making the dynamical convergence possible.

provides a survey of the conventional metrics for eddy activity as well as the finite-amplitude wave activity for selected model simulations. Section 5 reports the result of the wave activity budget, which leads to an explanation for the sensitivity behavior of the jet stream shown in Fig. 1, and demonstrates a probable convergence of the effective diffusivity, which is also indicative of the convergence of the governing dynamics of the jet stream. Last, we conclude the paper with a summary and discussion.

2. Models and experiments

Three different AGCMs, differing in both dynamical core and model physics, are utilized for the purpose of the current study. These are an aquaplanet version of the Model for Prediction across Scales–Atmosphere (MPAS-A, hereinafter simply MPAS) and the High-Order Methods Modeling Environment (HOMME), both with the physics parameterizations of the Community Atmosphere Model, version 4 (CAM4; Neale et al. 2010), and the Community Atmospheric Model, version 3 (CAM3), with a spectral Eulerian dynamical core (Collins et al. 2006). These simulations were performed as part of an effort to adopt a hierarchical evaluation framework to understand model sensitivity to dynamical core and model resolution (Leung et al. 2013). For each model, a set of experiments was conducted at four different resolutions: 240, 120, 60, and 30 km for MPAS; 220, 110, 55, and 28 km for HOMME; and T42, T85, T170, and T340 spectral truncation for CAM3 (corresponding to ~300, 150, 75, and 37.5 km at the equator, respectively), following the aquaplanet experiment (APE) protocol of Neale and Hoskins (2001) using the standard control SST boundary condition and equinoctial radiative condition.

The MPAS atmospheric dynamical core is built using a horizontal C-grid scheme configured on spherical centroidal Voronoi tessellations (Du et al. 1999; Ringler et al. 2008), which is amenable to local mesh refinement. When configured with variable resolution meshes, both the conservation properties (including energy, mass, tracer, and potential vorticity) and rate of convergence are unchanged (Ringler et al. 2011). However, in the current study, only the simulations with the quasi-uniform resolutions are analyzed. A more detailed description of the MPAS atmospheric dynamical core can be found in Rauscher et al. (2013), O'Brien et al. (2013), and Leung et al. (2013). HOMME, also known as CAM_SE (for "spectral element"), uses a continuous Galerkin spectral finite-element method (e.g., Taylor et al. 2007; Taylor and Fournier 2010). This class of CAM dynamical core was designed from inception for fully unstructured grids. It has been demonstrated to improve the computational scalability of CAM compared to the spectral Eulerian and finite volume dynamical cores (Evans et al. 2013). In addition, HOMME is known to produce accurate solutions for the atmospheric test-case problems (e.g., Taylor et al. 2007; Lauritzen et al. 2010). See Evans et al. (2013) for more details about the HOMME dynamical core. Both MPAS and HOMME simulations use the identical CAM4 physics with a 10-min physics time step and 26 vertical levels with a model top at 2.2 hPa. The analysis on these two model groups is all based on a 4-yr model output after discarding the spinup period.

CAM3 simulations are the same as those in Li et al. (2011a,b), conducted originally to study the sensitivity of precipitation and its extremes to horizontal resolution. CAM3 uses the standard Eulerian spectral transform dynamical core and standard CAM3 physics [see details in Collins et al. (2006)]. The standard operating mode for the Eulerian dynamical core requires decreasing time steps with increasing horizontal resolution to maintain numerical stability in accordance with the CFL criterion (Courant et al. 1928). Here to avoid the variation due to changing time steps, only data from integrations with 5-min time step (the time step that ensures the numerical stability for all four resolutions) are analyzed. The simulation periods are 8, 4, 2, and 2 years for the T42, T85, T170, and T340 resolutions respectively, after a 1-yr spinup period.

Compared to CAM3 physics, CAM4 physics represents a major stride of improvement, in particular in the representation of deep convection by including the effects of convective vertical momentum transport (Richter and Rasch 2008) and introducing a dilute, rather than an undilute, approximation in the plume calculation (Neale et al. 2008). As such, MPAS, HOMME, and CAM3 form a two-pronged comparison: comparing between MPAS and HOMME reveals the differences due to dynamical core; comparing between MPAS or HOMME and CAM3 reveals the differences due to both dynamics and physics. As shown in Fig. 1, the jets in the three models appear to converge toward different states, a likely consequence of different dynamical cores and different physics. Meanwhile, MPAS and HOMME (both using CAM4 physics) tend to simulate more poleward jet latitude compared to CAM3, reflecting the improvement in the global angular momentum budget by considering the convective momentum transfer in the CAM4 physics (Richter and Rasch 2008).

In all three models, when the horizontal resolution doubles, all the adjustable parameters in the physics suite are fixed at the standard values except that the ∇^4

horizontal diffusion coefficients are reduced to minimize the impact of the diffusion while ensuring the numerical stability.

3. Diagnostic framework

Since the focus of our diagnosis is the key characteristics of the eddy-driven jet, the budget is built for the eddy momentum/vorticity forcing for the jet stream. To this end, we start with the familiar vorticity equation in pressure coordinates (Holton 2004; Trenberth et al. 1998)

$$\frac{\partial q}{\partial t} + \mathbf{v}_\psi \cdot \nabla q = -\nabla \cdot (\mathbf{v}_\chi q) + \kappa \nabla^2 q, \tag{3.1}$$

where q is the vertical component of absolute vorticity, and the horizontal advection term has been split into the components by the rotational $\mathbf{v}_\psi$ and nonrotational component $\mathbf{v}_\chi$ of the horizontal wind, the latter being combined together with the vorticity stretching term into flux form $[-\nabla \cdot (\mathbf{v}_\chi q)]$ on the right-hand side of the equation. The resultant convergence of the vorticity flux by the divergent flow is exactly the Rossby wave source described in Sardeshmukh and Hoskins (1988) and Trenberth et al. (1998). Note that (3.1) is derived from the horizontal momentum equations with only the curvature terms and the terms involving the vertical velocity neglected following the scale analysis of Holton (2004, chapter 2.4) and no assumption of quasi-geostrophy is made. A Laplacian form of diffusion is assumed in (3.1) with a quasi-constant coefficient κ to remove the enstrophy at smallest scales. In the context of this diagnostic study, this term can be thought of parameterizing the effects of all diffusive processes for q, although the exact order of the hyperdiffusion in the AGCMs is not necessarily second order. Interpreting $-\nabla \cdot (\mathbf{v}_\chi q)$ as the forcing/wave source, (3.1) assumes a general form of a forced advection–diffusion problem for a tracer q.

For a tracer governed by the advection–diffusion problem, the finest scale of the tracer is set by the so-called Batchelor scale, at which the creation of smaller and smaller scales of the tracer by advection or stirring is halted by the diffusion. A simple scale analysis gives the expression for the Batchelor scale: $\delta = (\kappa L/U)^{1/2}$, where U and L are the velocity and length scales of the flow that carries out the stirring. For flows within the Batchelor turbulence regime introduced in Introduction, L/U stays constant, hence the apparent finest scale of the tracer filaments is predominantly determined by the value of κ.

The advection–diffusion problem is one of the most recurrent problems in atmospheric and oceanic fluid dynamics. What is particularly interesting about this problem is that it can be mathematically converted to a one-dimensional forced-diffusion problem through area-coordinate transformation. The advantage of the transformation is that the interplay between the chaotic stirring by the resolved flow and the subgrid micro-diffusion can be condensed into one single parameter: effective diffusivity (Nakamura 2008). The resultant streamwise depiction of tracer fits nicely the need of the current investigation as we link the intensity and the location of upper-level jet stream to the magnitude and the location of the absolute vorticity gradient maximum, which is strongly shaped by the profile of K_{eff} (Nakamura 2008). Specifically, the area-coordinate transformation is realized through the integral operator $\Delta\mathcal{A}$ defined as (Nakamura and Zhu 2010; Chen and Plumb 2014)

$$\Delta\mathcal{A}(\ ) \equiv \frac{1}{2\pi a \cos\phi_e}\left[\iint_{q>Q} (\ )\, dS - \iint_{\phi>\phi_e} (\ )\, dS\right], \tag{3.2}$$

where Q demarcates the value of a wavy, instantaneous tracer (absolute vorticity q in our case) contour with an equivalent latitude ϕ_e. The equivalent latitude ϕ_e for a given Q value (upper case is used to emphasize its Lagrangian property) is determined uniquely by the requirement that the area enclosed by the Q contour toward the polar cap equals the area poleward of latitude ϕ_e; that is,

$$\iint_{q>Q,\phi<\pi/2} dS = \iint_{\phi_e<\phi<\pi/2} dS = 2\pi a^2(1-\sin\phi_e), \tag{3.3}$$

where a is the radius of Earth. The net effect of the integral operator (3.2) is the difference of the integration of the quantity in the parentheses over the red areas minus that over the blue areas illustrated in Fig. 2. Applying $\Delta\mathcal{A}$ to q, we can define the finite-amplitude wave activity (FAWA) based on the absolute vorticity

$$\begin{aligned} A(\phi_e, t) &\equiv \Delta\mathcal{A}(q) \\ &\equiv \frac{1}{2\pi a \cos\phi_e}\iint_{q>Q,\phi\le\phi_e} q\, dS - \iint_{q<Q,\phi>\phi_e} q\, dS. \end{aligned} \tag{3.4}$$

The desirable properties of FAWA are manifold. First, $A(\phi_e, t)$ is positive definite and its dissipation by irreversible mixing is always negative. Second, for inviscid, conservative flow (i.e., no wave source and irreversible mixing), the nonacceleration theorem holds exactly; that is,

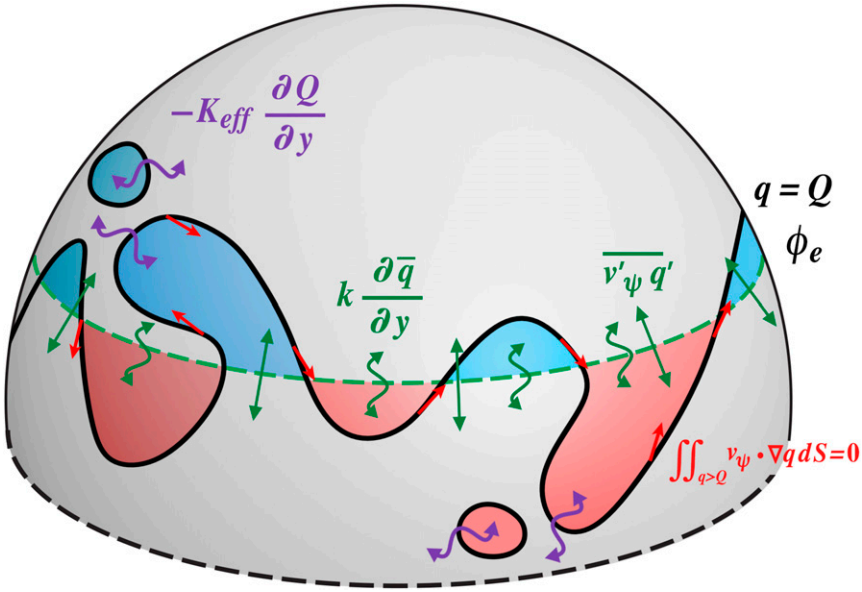

FIG. 2. A schematic illustrating the area-coordinate transformation and the terms constituting the finite-amplitude wave activity budget [see (3.6)]. The terms participating in the budget are (i) flux through the parallel of the latitude ϕ_e (straight green arrows), (ii) diffusive flux across the latitude ϕ_e (wavy green arrows), and (iii) diffusive flux across the contour of Q (wavy purple arrows). The advective flux through the Q contour (red arrows) can be shown to be identically 0 through the divergence theorem. See section 3 for the definition of the wave activity and the area operator for the area-coordinate transformation.

$$\frac{\partial}{\partial t}(\overline{u} + A) = 0, \tag{3.5}$$

for both small- and finite-amplitude wave perturbations. Compared to the traditional Eulerian definition of wave activity (cf. Andrews et al. 1987, section 3; Vallis 2006, section 12.1), (3.5) does not contain the divergence of wave activity flux as it has been absorbed in the movement of the Q contour. Third, the forced advection–diffusion problem as depicted by (3.1) can be transformed through applying the integral operator $\Delta\mathcal{A}$ to a one-dimensional problem in the equivalent latitude ϕ_e (Nakamura and Zhu 2010; Chen and Plumb 2014):

$$\frac{\partial A}{\partial t} + \overline{v'_\psi q'} = \Delta\mathcal{A}[-\nabla\cdot(\mathbf{v}_\chi q)] - K^e_{\text{eff}}\frac{\partial Q}{\partial y}. \tag{3.6}$$

Each term in (3.6) corresponds exactly to that in (3.1). The derivation of the first term is straightforward from the definition of the wave activity. The second term on the left results from the area-coordinate transformation of the second term on the left-hand side of (3.1), representing the zonal mean vorticity flux by the rotational component of the meridional wind. The second term on the right is derived from the area-coordinate transformation of the second term on the right-hand side of (3.1), indicating the dissipation of wave activity in the form of downgradient diffusion of absolute vorticity. The effective eddy diffusivity is defined as (Chen and Plumb 2014)

$$K^e_{\text{eff}} \equiv K_{\text{eff}} - \kappa\frac{\partial\overline{q}/\partial y}{\partial Q/\partial y}, \tag{3.7}$$

somewhat differing from the effective diffusivity first defined by Nakamura (1996). It represents the enhancement of diffusivity associated with the diffusive flux across the tracer contour Q through stirring that stretches the interface between fluids of different Q and sharpens the local tracer gradient (the corresponding dissipation is $K_{\text{eff}}\partial Q/\partial y$, as indicated by the purple wavy arrows in Fig. 2) minus the regular diffusivity across the parallel ϕ_e (the corresponding dissipation is $\kappa\partial\overline{q}/\partial y$, as indicated by the green wavy arrows in Fig. 2). As will be shown later, K_{eff} is much larger than κ by several orders of magnitude and therefore the first term in (3.7) dominates the diffusive process. As a result, all the discussion about K_{eff} hereafter can be equally applied to K^e_{eff}. See Allen and Nakamura (2001) or Nakamura and Zhu (2010) for the detailed derivations of K_{eff}. For the convenience of the readers, the derivations of both terms are provided in appendix B. In summary, (3.6) provides a diagnostic and interpretation framework for the eddy vorticity flux (which drives the eddy-driven jet directly) in the simulations with

different AGCMs at varying resolutions. We caution that the exact values of $K_{\rm eff}$ can differ depending on the specific tracer (absolute vorticity versus PV) and the vertical coordinate (isobaric versus isentropic) on which the diagnosis is carried out. In the current investigation, the budget analysis will be performed for the absolution vorticity-based wave activity at 250-hPa isobaric level.

Using a somewhat different integral operator, Nakamura (1996) showed that (3.1) can be transformed to be

$$\frac{\partial Q}{\partial t} = \frac{\partial}{\partial y}\left(K_{\rm eff}\frac{\partial Q}{\partial y}\right) - \frac{\partial}{\partial y}\{\Delta\mathcal{A}[-\nabla\cdot(\mathbf{v}_\chi q)] + \overline{X}\}, \quad (3.8)$$

where $\overline{X}$ represents the zonal-mean frictional force. It can also be readily derived from taking a y derivative of the result of adding the wave activity (3.6) to the corresponding equation for the zonal-mean zonal wind $\partial\overline{u}/\partial t = \overline{v'_\psi q'} + \overline{X}$ and making use of the relationship between the Lagrangian and Eulerian-mean vorticity

$$\frac{\partial Q}{\partial y} = \frac{\partial \overline{q}}{\partial y} - \frac{\partial^2 A}{\partial y^2} = \beta - \frac{\partial^2(\overline{u}+A)}{\partial y^2}.$$

What is remarkable about (3.8) is that for a given wave source, the state of Q is entirely determined by $K_{\rm eff}$. By corollary, if the numerical model converges on the value of $K_{\rm eff}$, it also converges on the solution of Q, to the extent that the wave source itself does not change with resolution (in practice, the wave source can still vary with resolution).

As shown in appendix B (see also Nakamura 1996; Allen and Nakamura 2001), the effective diffusivity takes the form

$$K_{\rm eff} = \kappa\frac{L_e^2}{L_0^2}, \quad \text{where} \quad L_e^2 = L_0^2\left[\frac{a^2\langle|\nabla q|^2\rangle_Q}{(\partial Q/\partial\phi_e)^2}\right] \quad \text{and}$$
$$L_0 = 2\pi a\cos\phi_e. \quad (3.9)$$

Here, L_e is the equivalent length and it reduces to the actual perimeter length of the contour when $|\nabla q|$ is constant on the contour. The subscripted angle bracket denotes the average along the Q contour. The squared equivalent length normalized by its minimum possible value L_0 measures the magnifying factor on the microscale diffusion by the stretching of the contour. Note that L_e is dependent strongly on resolution, increasing monotonically with finer grid scale (see Fig. 10b).

Given this monotonic dependence of L_e on resolution and the arbitrariness in the form of the subgrid-scale diffusion and the choice of diffusivity coefficient in AGCMs, it is quite a surprise that $K_{\rm eff}$ asymptotes to a quantity independent of κ under the large Pe limit. This can be demonstrated through the following scaling analysis for the type-II flow described in the introduction (see also Shuckburgh and Haynes 2003; Marshall et al. 2006). In a mixing region characterized by a stretching rate $S = U/L$ with an area of $A_m \sim L^2$, the mixing region will be filled with tracer filaments with thickness of the Batchelor scale $\delta = (\kappa/S)^{1/2}$. For small enough κ (so Pe $\equiv SL^2/k$) that the area of the mixing is largely conserved, the typical length of the filaments is

$$L_e \sim \frac{A_m}{\delta} \sim A_m\left(\frac{S}{\kappa}\right)^{1/2}. \quad (3.10)$$

This scaling of equivalent length with stretching rate is consistent with the impression that L_e is usually large in regions of strong stirring and small in regions of weak stirring. It follows that

$$K_{\rm eff} = \kappa\frac{L_e^2}{L^2} \sim \frac{\kappa A_m^2}{L^2}\frac{S}{\kappa} \sim \frac{SA_m^2}{L^2} \sim SA_m \sim UL. \quad (3.11)$$

Thus, for flows in the large Pe regime, $K_{\rm eff}$ is determined by the length and velocity scales of the stirring flow, independent of κ.[2] In addition, if the stretching rate S is fixed, scaling above (or $K_{\rm eff} \sim SL^2$) would suggest a fairly fast convergence of $K_{\rm eff}$ at a k^{-2} rate [k denotes the wavenumber of the eddy kinetic energy (EKE) spectrum] as the scale of stirring decreases. This relatively fast convergence of $K_{\rm eff}$ bestows a counterintuitive property on the tracer fields controlled by the chaotic advection/stirring: the streamwise depiction of the evolution of the tracer behaves as if being diffused via coefficient $K_{\rm eff}$, irrespective of the magnitude of κ. The corresponding governing equation for absolute vorticity as a tracer on equivalent latitude is expressed by (3.8).

The insensitivity of $K_{\rm eff}$ to κ is especially appealing in view of the fact that the real value of κ is difficult to estimate accurately for the real atmosphere and ocean (Vallis 2006, section 10) and is presumed to be much smaller than that typically used in numerical models (e.g., Thuburn 2008). In numerical models, as the horizontal resolution improves and allows smaller and smaller κ, the simulation approaches closer and closer to the large Pe regime and $K_{\rm eff}$ converges toward its real-world

[2] In flows that are not characterized by a stretching rate fixed over all scales [e.g., three-dimensional turbulent flows or more generally the nonsmooth flows discussed by Falkovich et al. (2001)], the relevant S will increase with decreasing L and no simple relations between L_e and S or between $K_{\rm eff}$ and S exist.

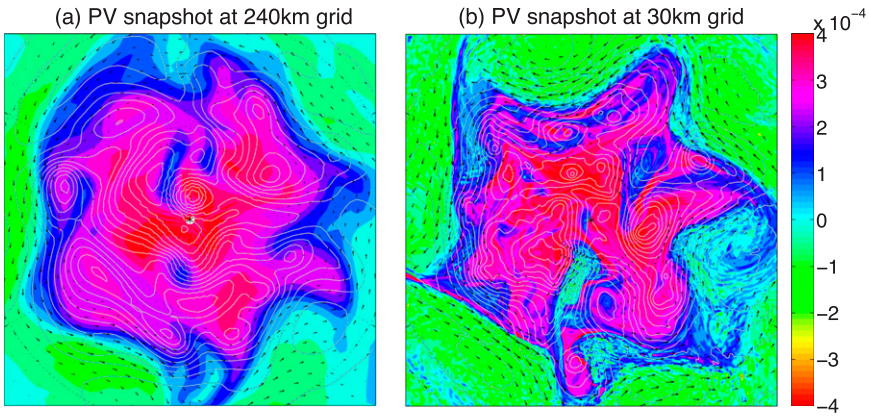

FIG. 3. Snapshots of PV (shadings; s^{-1}), velocity vectors (arrows; $m\,s^{-1}$) at the 250-hPa level, and sea level pressure (contours; hPa) at (a) 240- and (b) 30-km MPAS. The maximum winds in (a) and (b) are 70 and 109 $m\,s^{-1}$, respectively.

counterpart. Insofar as K_{eff} converges to its true value, the modeled Q converges toward the "true" solution of the respective model. Promising examples of convergence were demonstrated for the effective diffusivity in the Southern Ocean (Marshall et al. 2006) and in idealized chaotic advection by a simple analytic flow (e.g., Shuckburgh and Haynes 2003). As will be shown in section 5, in three AGCMs examined herein K_{eff} shows signs of convergence as the resolution increases from the second highest to the highest resolutions. This arguably offers the dynamical underpinning for the apparent converging behavior of the eddy-driven jet in these AGCMs.

4. Survey of basic statistics

a. Comparison of snapshots

Figure 3 compares snapshots of upper-level PV, upper-level wind, and sea level pressure simulated by MPAS at 240- and 30-km resolutions. Note that these two snapshots are picked randomly to illustrate the characteristic differences due to resolutions; the synoptic details are not comparable between them. At first glance, one can see clearly a wavenumber-6 component in the PV and flow fields in both cases. As will be shown next, this represents the typical scale of wave activity injection from baroclinic instability. Clearly, the PV snapshot at 30-km resolution exhibits many more details than at 240 km. For instance, the dipolar structure in the SLP at the middle front of the domain with a cutoff low to the south and a blocking to the north is associated with entrainment of low PV to the center of the blocking and shedding of high PV over the low pressure center. To the west of this structure, a remarkable elongated thin PV filament (or streamer) stretches at least 6000 km long in a northeast–southwest direction. Correspondingly near the surface, there exists an intense stream of northeastward moisture transport along the southeast edge of the surface trough (not shown)—a typical characteristic of atmospheric rivers. It is also noticeable that the pressure gradient and thus the associated geostrophic wind at the surface is stronger in the 30-km snapshot than the 240-km one. The filamentation of the PV contours and the associated sharpness in many meteorological fields are difficult for the coarse model grid to capture. One obvious reason is that the coarse-resolution model is overly diffusive (the diagnosed midlatitude κ is about 2.5×10^5 versus $8 \times 10^3\,m^2\,s^{-1}$ estimated for the case of 30-km resolution), destroying the fine structures in PV. Relatedly, the maximum wind in Fig. 3b is 109 $m\,s^{-1}$, approximately 55% greater than the maximum wind speed in Fig. 3a.

Note that, however, the climatological mean zonal wind is weaker in the 30-km resolution than that in the 240-km resolution (Fig. 1). Meanwhile, the difference in the spatial scales of the wind field is much harder to discern between model resolutions. As to be shown next, the wave activity is a much more discriminative metric than eddy kinetic energy. Another important feature to note in Fig. 3 is that the upper-level winds tend to flow along the isolines of the PV at upper levels. This reflects the material nature of PV, namely that the advection of a material tracer across a given tracer contour by the rotational component of the wind is identically zero (as illustrated in Fig. 2; see appendix A for the proof). It should be stressed that all the properties discussed about the PV can be equally applied to absolute vorticity as well.

Before getting to more quantitative analysis, just by eyeballing the two snapshots one can see that (i) the smallest scale visible in the simulation is much finer in the high resolution than the coarse resolution. Were the

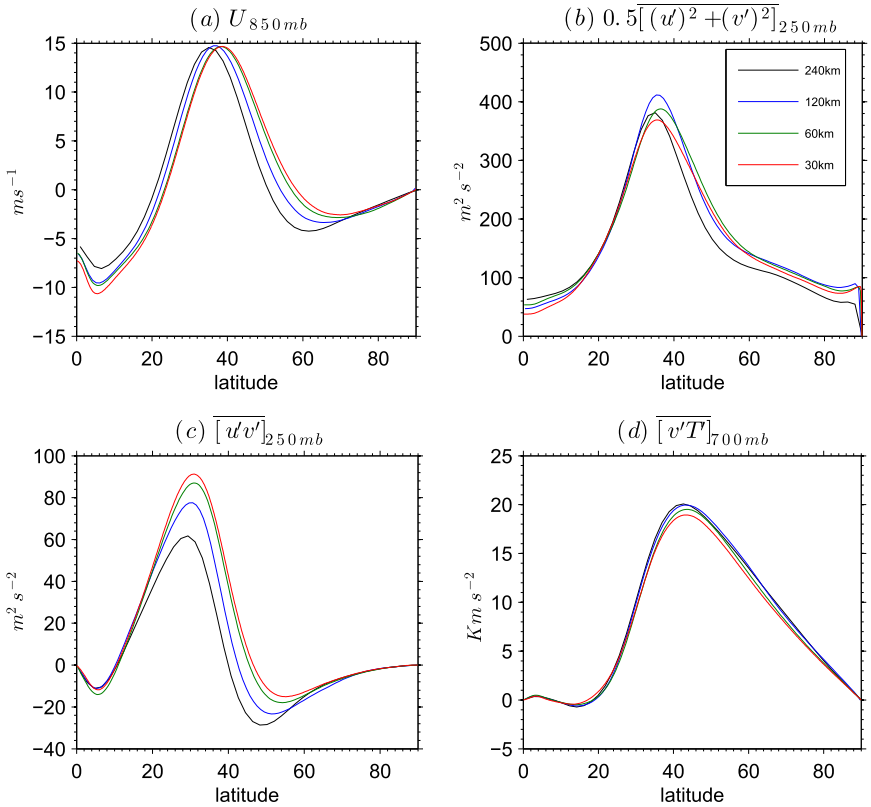

FIG. 4. Conventional eddy statistics in the MPAS aquaplanet simulations at 240- (black), 120- (blue), 60- (green), and 30-km (red) resolutions: (a) 850-hPa zonal-mean zonal wind (m s^{-1}), (b) 250-hPa eddy kinetic energy (m^2 s^{-2}), (c) 250-hPa eddy momentum flux (m^2 s^{-2}), and (d) 700-hPa eddy poleward heat flux (K m s^{-1}).

flows residing in the Batchelor regime, the scale difference would mainly reflect the differences in subgrid-scale diffusion between the two snapshots; (ii) the PV contours (or isolines of any tracer) have greater complexity in the high resolution than the coarse one and accordingly the equivalent length in the former is expected to be larger; (iii) the sharpness of PV gradient in the 30-km resolution also leads one to speculate that the effective diffusivity therein might be smaller; and (iv) the upper-level westerly jet tends to be positioned more poleward in the 30-km resolution compared to the 240-km one. These observations will be quantified using the diagnostic framework developed in section 3.

b. Comparison of basic eddy statistics

Since the MPAS aquaplanet experiments archive the most complete set of variables at 6-hourly frequency, the analysis in this subsection will be focused on the MPAS model. Nevertheless, the conclusions reached here can be generalized to other aquaplanet AGCMs, unless stated otherwise.

We begin with the conventional metrics for mean circulation and eddy activity. Figure 4a shows profiles of 850-hPa zonal-mean zonal wind. With increasing resolution, the peak of the near surface wind shifts poleward progressively, whereas the shift appears to come to a halt at the last resolution doubling—a suggestion that the jet position has converged. The same convergence behavior is also seen in the CAM3 and HOMME simulations (e.g., Fig. 1). Since the near-surface westerlies are eddy driven, their convergence is the result of the convergence of the eddy momentum flux, just as what Fig. 4c depicts. Figure 4c also shows that the magnitude of eddy momentum flux increases monotonically with increasing resolution. On the other hand, there is no systematic relationship between the eddy kinetic energy and the resolution (Fig. 4b). Moreover, one cannot attribute the increase of eddy momentum flux to the increase of the eddy heat flux either, as the eddy heat flux at lower troposphere tends to decrease with resolution (Fig. 4d). As to be shown later, this discrepancy in sensitivity between the upper-level eddy momentum (vorticity) flux and the low-level eddy heat flux can be reconciled by the larger effective diffusivity in the coarse-resolution model, wherein less wave activity survives the dissipation to reach the free propagating

level despite the greater wave activity source from below.

c. Wave activity and spectrum

Figure 5 summarizes the 250-hPa wave activity as a function of equivalent latitude computed using (3.4) for the FAWA for all the simulations with all the three models. The vertical and meridional distributions of wave activity and their seasonal dependence computed from reanalysis data were described in Nakamura and Solomon (2010). Here to succinctly compare the wave activity from 12 model simulations, only the wave activity at the 250-hPa level is shown in Fig. 5. Compared to the EKE at the same level (Fig. 4b), the wave activity peaks at much higher latitudes. Different from EKE (cf. Fig. 4b), wave activity shows acute sensitivity to the resolution: higher resolution allows for substantially larger magnitude of wave activity. This should be expected as higher resolution can resolve finer details of the PV or vorticity contours, whereas EKE density drops fast with wavenumber and resolving the energy at smaller scales does not boost the total EKE much.

The monotonic relationship between the wave activity level and the resolution is clear in CAM3 and MPAS; but in the HOMME group, 55-km resolution is an outlier, showing the largest wave activity level among all the simulations. Similar "breaking ranks" behavior of the 55-km HOMME was also found in the eddy momentum flux, jet maximum speed, water vapor content (Hagos et al. 2015), and equatorial precipitation (Yang et al. 2014). This aside, wave activity magnitude is generally higher in CAM3 and HOMME than MPAS for the equivalent grid resolution, consistent with the experience that MPAS has a more diffusive dynamics compared to HOMME and CAM Eulerian dynamical cores (M. Taylor 2014, personal communication).

It is also informative to compare the wave activity spectrum with the corresponding EKE spectrum. Figure 6 depicts the wave activity spectrum (solid lines) together with the EKE spectrum (dashed lines) for the latitudinal range centered about the peak of the EKE (35°–45°) at the 250-hPa level. This is the first time that wave activity spectrum has ever been computed and shown. The wave activity spectrum cannot be computed directly from the total tracer fields using Fourier transform method. Instead, the tracer fields are truncated in the zonal direction for each wavenumber from wavenumber 1 to the largest wavenumber that the resolution allows and the accumulative wave activity is computed for each of the truncations. The wave activity spectrum density at certain wavenumber, say k, is then derived by differencing the accumulative wave activity

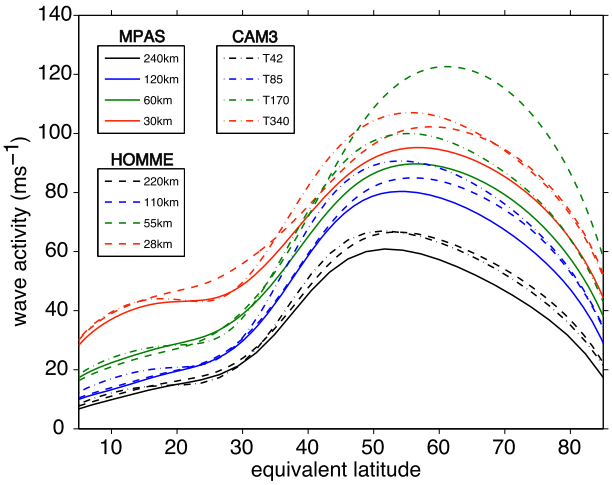

FIG. 5. The 250-hPa finite-amplitude wave activity (m s^{-1}) simulated by three AGCMs, each running at four different resolutions.

between wavenumbers k and $k - 1$. To facilitate the comparison with the wave activity spectrum, which is a function of the zonal wavenumber k, we use Fourier transform, instead of the spherical harmonics expansion, to compute the EKE spectrum for each latitude grid. As such, the resultant EKE spectrum is also a function of zonal wavenumber.

Several features of the EKE spectrum differing from the conventional spherical harmonics spectrum are noteworthy. First, the spectra here for all four resolutions exhibit a clear local peak in the synoptic range (wavenumbers 4–12, corresponding to 9000–3000 km; see also Tung and Orlando 2003) centered at wavenumber 6, representing the wave energy and enstrophy injection due to baroclinic instability. For scales immediately smaller than the synoptic range, the spectral slope is characterized by the classic -3 slope of the inertial range where the forward enstrophy cascade is dominant (e.g., Kraichnan 1967; Lindborg 1999). The spectrum of the 30-km resolution shows a hint of flattening near wavenumbers 50–60 (~600 km), while exhibiting no sign of Nastrom–Gage transition from -3 to $-5/3$. We did compute the global spherical harmonics spectra for these MPAS simulations and the result is typical of those found in many other similar aquaplanet experiments (e.g., Williamson 2008; Evans et al. 2013) as well the observation (Nastrom and Gage 1985), with an evident transition from -3 to $-5/3$ at approximately 600 km. Looking at other latitudes, we found that the transition from -3 to $-5/3$ appears to originate from the tropical spectrum (not shown), whereas the spectrum of high latitudes has its -3 slope extending all the way to the dissipation scales. The underlying cause for the different characteristics of the EKE spectra at

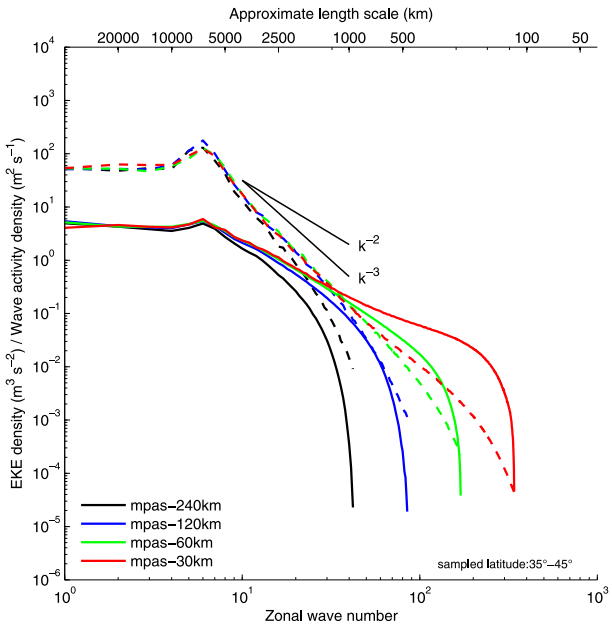

FIG. 6. Fast Fourier transform spectra of finite-amplitude wave activity (solid) and EKE (dashed) sampled around the jet latitudes (between 35° and 45°) in the simulations of MPAS. A −3 slope and a −2 slope are drawn to guide the eyes.

different latitudes is out of the scope of the current investigation.

The wave activity spectrum shows a much flatter slope than the corresponding EKE spectrum, reflecting the fact that FAWA is much more sensitive to the small scales than EKE. As such, FAWA is a much more discriminative metric for discerning the small scales and distinguishing the effect of resolution. In the forward enstrophy cascade range, the slope of wave activity spectrum is roughly half of that of the EKE spectrum, with the best fit of slope at −1.7. It is yet to be understood why the inertial range of the wave activity has a $k^{-1.7}$ spectrum. Dimensional analysis of the wave activity implies that it should scale as $A \sim (U/L)L_{\mathrm{eq}}$. If the stretching rate in two-dimensional turbulence is independent of the scale, the −1.7 slope should mainly reflect the sensitivity of the equivalent length to eddy scale. Hypothetically, if L_{eq} scaled linearly with L, the wave activity would have the same spectrum as velocity, characteristic of a −2 slope (plotted in Fig. 6 for reference). As L_{eq} matters more to the effective diffusivity than EKE, the much weaker dependence of wave activity spectrum on eddy scale implies that to capture the scales needed for the convergence of the effective diffusivity, models need to resolve a much broader inertial range than deemed necessary from the EKE convergence.

Characteristically identical wave activity spectra are also obtained for CAM3 model (not shown), as the four spectral truncations (T42, T85, T170, and T340) of CAM3 have similar grid spacing to the corresponding four resolutions of MPAS. All in all, wave activity is a more discriminative metric than the traditional metrics for the midlatitude disturbances and it might present an advantage for quantifying model convergence or the lack thereof.

5. Results of wave activity budget

a. *Mechanisms for the jet sensitivity to resolution*

Before getting to the result of the wave activity budget, we air a note of caution that budget analysis can only help reveal a dynamically consistent picture but cannot address the issue of causation. Nevertheless, through this exercise we can advance our understanding by pinpointing the key factors and processes involved in the sensitivity behavior of the westerly jet.

The wave activity budget is conducted at the 250-hPa level following (3.6) to account for the meridional profiles of the eddy forcing (i.e., eddy vorticity flux) for the eddy-driven jet and its sensitivity to resolution. The results for MPAS, CAM3, and HOMME are shown in Figs. 7–9, respectively. First of all, as defined and formulated in section 3, the wave activity and its source are always positive, the Lagrangian vorticity (Q) in equivalent latitude is monotonic with latitude (in contrast to the zonal-mean vorticity $\overline{q}$), and the dissipation of wave activity is negative, representing downgradient diffusion of the Lagrangian vorticity. The eddy vorticity flux peaks at the latitude of the jet stream, driving the eddy-driven component thereof. The positive eddy vorticity flux near the jet is clearly up the gradient of both the Eulerian-mean and Lagrangian absolute vorticity (Fig. 7f), a feature distinct from the eddy PV flux in the upper troposphere (see Vallis 2006, ch. 12).

Consistent with the profiles of the momentum flux (Fig. 4c), the eddy vorticity flux increases in magnitude and its peak shifts poleward as resolution increases, with a suggestion of convergence between the second highest and the highest resolutions (Fig. 7a). The same can also be said to the result for the CAM3 simulations (Fig. 8a). Part of the sensitivity of the magnitude of the eddy vorticity flux can be attributed to the increase of wave activity source near the jet core with increasing resolution (Figs. 7b and 8b). However, despite the enhanced eddy activity source, the near-surface westerly wind shows little change in magnitude with resolution (Fig. 4a); the upper-level jet instead decreases in

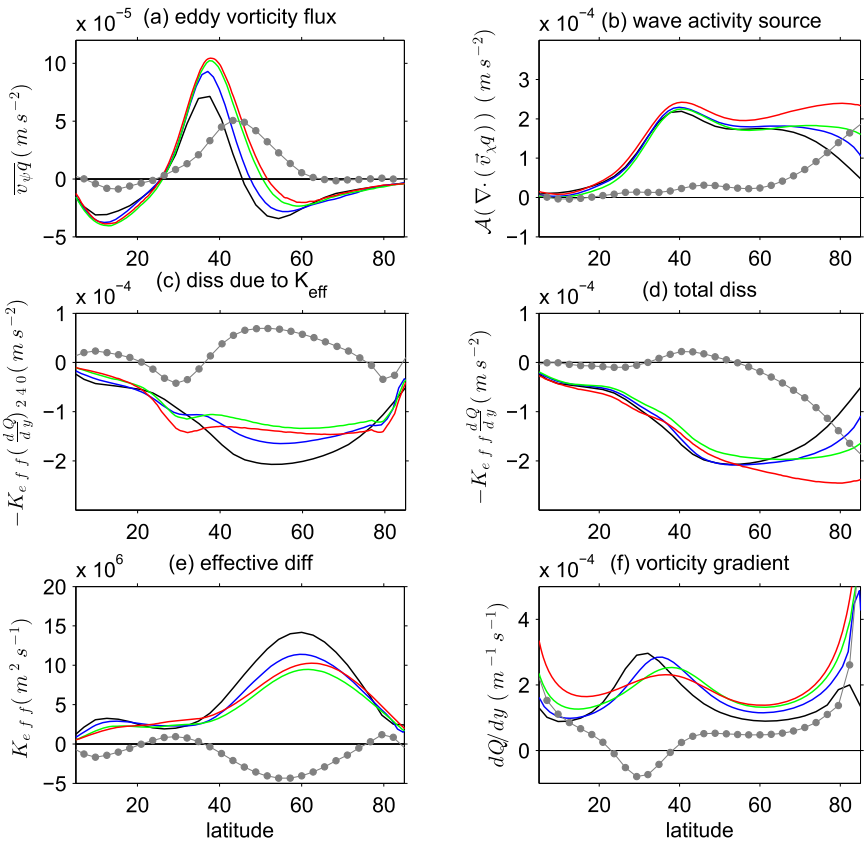

FIG. 7. Result of wave activity budget at 250 hPa for the equilibrium MPAS simulations and the result the contrast between the highest and lowest resolutions: (a) eddy vorticity flux, (b) wave activity source, (c) dissipation with dQ/dy fixed at the 240-km condition representing the dissipation due to the change of K_{eff}, (d) total dissipation, (e) effective diffusivity K^{e}_{eff}, and (f) dQ/dy. In each panel, the difference between the highest and lowest resolutions is displayed as dotted gray line.

intensity with resolution (Fig. 1). This suggests that the dissipation process might be more important in determining the upper-level jet intensity. Indeed, the peak Lagrangian vorticity gradient, an alternative indicator for the jet intensity, at the 250-hPa isobaric level decreases in magnitude monotonically with increasing resolution (Fig. 7f). As will be demonstrated next, this sensitivity of jet intensity can be largely attributed to the change of effective diffusivity.

The near-surface westerly wind is the result of the momentum balance between the vertically integrated eddy vorticity flux and the near-surface friction. The fact that the surface westerly wind intensity stays roughly the same despite the eddy vorticity flux increases with resolution implies that the surface friction should increase with resolution as well (which is possible for a nonlinear bulk formula for surface stress).

Comparing between the highest and the lowest resolutions, the eddy vorticity flux difference is characterized by an increase at the poleward flank and a weaker decrease at the equatorward flank of the mean flux peak of the 240-km case (Fig. 7a, gray line), driving the poleward shift of the surface westerlies in the 30-km resolution relative to the 240-km resolution. By the formulation of the budget, the eddy vorticity flux difference is maintained by the difference of wave activity source (Fig. 7b) and that of dissipation of wave activity (Fig. 7d). The former difference can be related to the reduced effective diffusivity in the 30-km case wherein the waves coming from below and passing through the dissipation region are less damped and hence more wave activity can reach the propagation levels (approximately between 150 and 350 hPa), giving rise to greater wave source there. The dipolar structure, on the other hand, comes mostly from the difference in dissipation, a term that can be further decomposed into the change due to the change of the eddy effective diffusivity but with the absolute vorticity gradient fixed

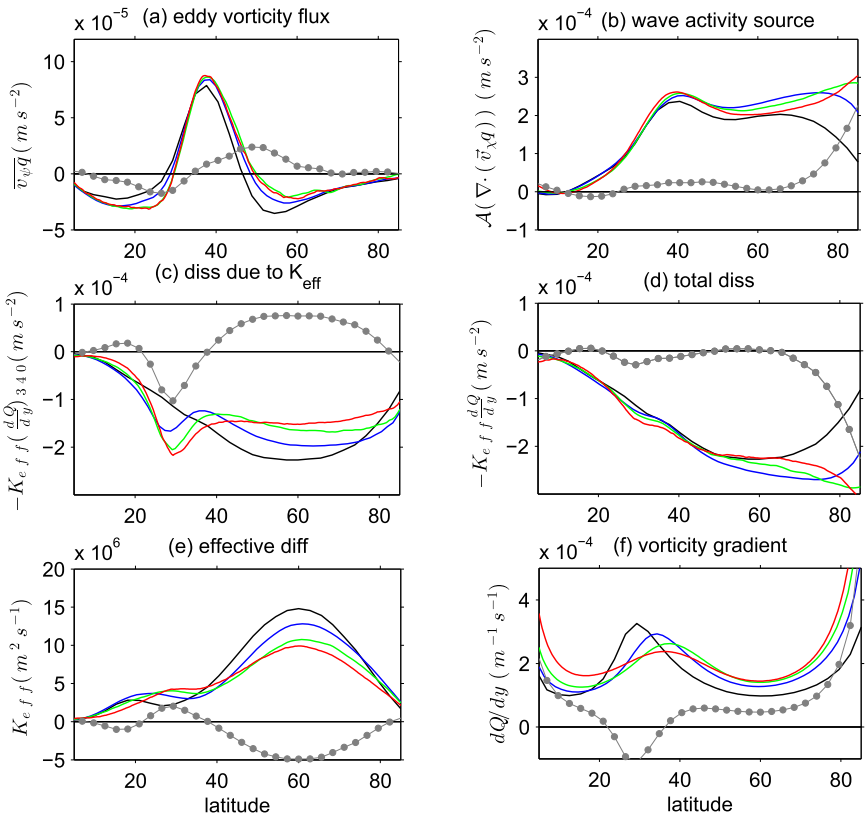

FIG. 8. As in Fig. 7, but for CAM3.

at 240-km resolution [i.e., $K^e_{\rm eff}(\partial Q/\partial y)_{240}$, Fig. 7c] and that due to the change of vorticity gradient (Fig. 7f). Apparently, the former is characteristic of a dipole and plays a greater part in maintaining the poleward shift in the wave vorticity flux.

From the analysis above, effective diffusivity appears to hold the key to understanding the sensitivity of the wave activity flux and dissipation to resolution. Increased horizontal resolution reduces the effective diffusivity and the concomitant dissipation of wave activity over the wave source region; more wave activity that has survived the dissipation propagates equatorward in the upper level so as to produce greater poleward momentum flux and hence greater eddy momentum forcing on the mean wind at the poleward flank of the jet. Since the equatorward-propagating waves will break and end up being absorbed at the critical layer at the equatorward flank of the jet, it is possible that the greater dissipation in the subtropics is the result of the larger equatorward propagation of wave activity. In the meantime, higher resolution might also contribute to the increase of the subtropical effective diffusivity by resolving more details of the filamentation of the vorticity and hence large equivalent length at the latitudes of the wave breaking (see Fig. 10b), even with the same influx of wave activity. This enhanced dissipation occurs at around 30° latitude (see Fig. 7c), where the angular momentum is constantly replenished by the poleward momentum advection by the upper branch of the Hadley cell. This angular momentum source tends to be more heavily damped in the higher-resolution simulation due to the greater effective diffusivity there (see also Figs. 8c,d for CAM3), a tendency consistent with the weaker upper-level jet stream associated with higher resolution.

Since the sensitivity of the jet stream is rooted in the magnitude and meridional structure of the effective diffusivity, its relationship to resolution warrants further examination. It should be noted here that $K^e_{\rm eff}$ is estimated as the residual of (3.6) and κ as $K^e_{\rm eff}/(L_e^2/L^2)$ as all other terms can be explicitly calculated from the model output. Still we can place considerable confidence in the inferred $K^e_{\rm eff}$ since little assumption (say, no quasigeostrophic approximation) has been made in deriving (3.8). Figure 10 shows the profiles of the diffusion coefficient κ (Fig. 10a) and those of the normalized equivalent lengths (L_e^2/L_0^2, Fig. 10b) computed directly from (3.9). One can see that both quantities are acutely sensitive to the resolution, with κ decreasing and L_e^2/L_0^2

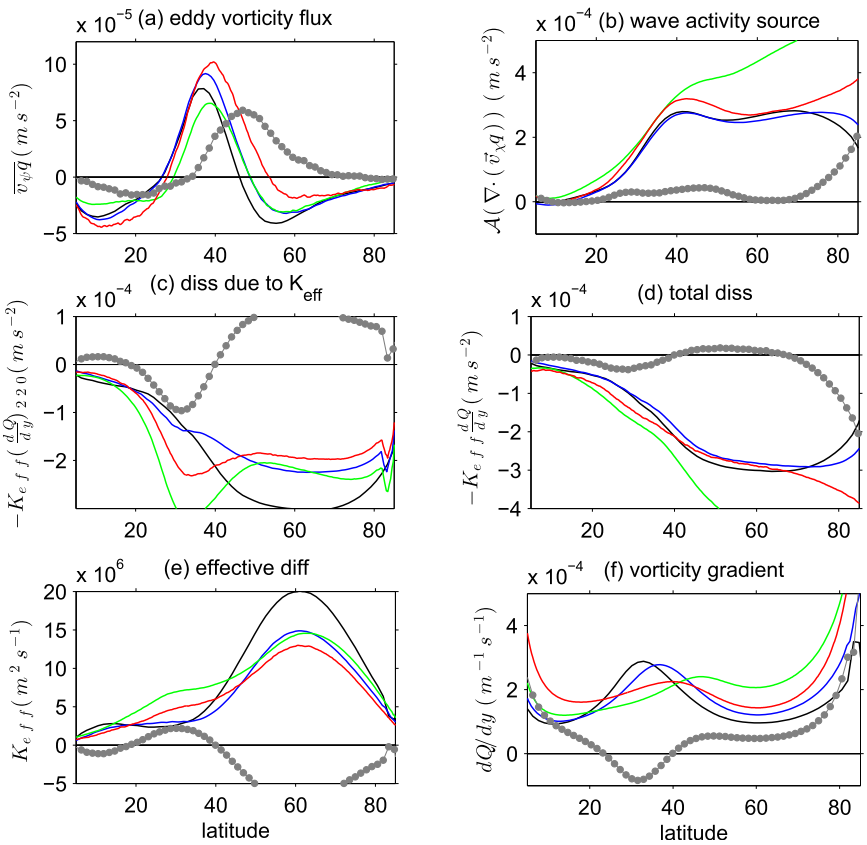

FIG. 9. As in Fig. 7, but for HOMME.

increasing sharply with resolution (more so in the tropics than extratropics). Effective diffusivity K_{eff} is the product of the two and shows much less sensitivity to resolution due to their compensation (Fig. 7e). Still, K_{eff} generally decreases with increasing resolution, especially in the extratropics.

Equivalent length has been widely used for diagnosing the irreversible mixing and diffusive flux in the stratosphere and upper troposphere (e.g., Allen and Nakamura 2001; Haynes and Shuckburgh 2000a,b). Although idealized, the meridional profiles of equivalent length in the aquaplanet MPAS (Fig. 10b) resemble in spirit that observed in the upper troposphere and lower stratosphere (e.g., plate 2 in Haynes and Shuckburgh 2000b). In particular, the minimum in the subtropics corresponds well to the upper-level jet that acts as a barrier for mixing. In addition, this minimum moves progressively poleward as does the jet with increasing resolution. More importantly, the variation of the equivalent length with latitude seems to be dominating the profiles of effective diffusivity K_{eff}, whereas κ assumes a much smoother function of latitude. A clarification of the meaning of κ is in order: when applied to full AGCMs, it represents not only Laplacian or higher-order diffusion, but also other implicit forms of diffusion resulting from the numerics, gravity wave drag, and convection. Notwithstanding, the asymptotic behavior of K_{eff} in the large Pe number regime is constrained by the smooth nature of the two-dimensional turbulence and should be insensitive to the exact form of the diffusion and the corresponding coefficient.

All the discussion above can be equally applied to the simulations of CAM3 (Fig. 8). The 220-, 110-, and 28-km resolutions of HOMME show similar sensitivity (Fig. 9) in their wave activity budget to MPAS and CAM3, but with the 55-km case behaving as an outlier. It has the largest wave activity source, largest total dissipation, smallest positive vorticity flux, and largest subtropical effective diffusivity of the group (green line in each panel of Fig. 9). The reason for this outlier behavior of the 55-km HOMME is unclear, a subject for future investigation.

b. *Importance of K_{eff} in the jet sensitivity to resolution*

Since the westerly jet can be thought of as the manifestation in momentum of the maximum dQ/dy

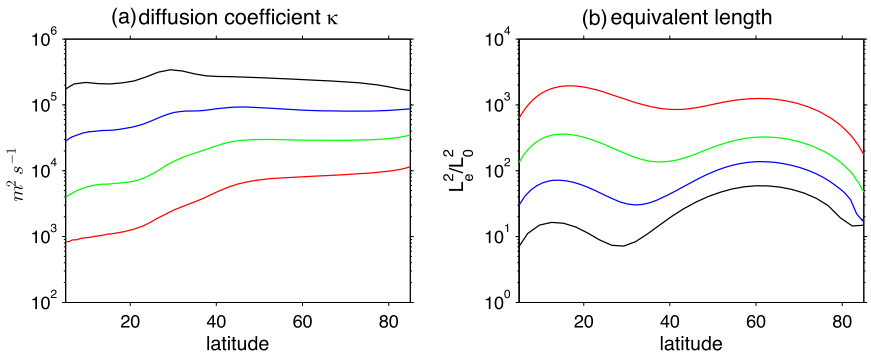

FIG. 10. (a) Diffusion coefficient κ and (b) normalized equivalent length in MPAS simulations. The color convention follows that of Fig. 7.

(and indeed they show the same sensitivity to resolution), the sensitivity of the former to resolution can be translated into that of the latter through the dynamical equivalence between (3.6) and (3.8). In (3.8) if we ignore the friction term, which is generally small in the upper troposphere, the equilibrium balance is then maintained between the gradient of diffusion and that of the wave source:

$$\frac{\partial}{\partial y}\left(K_{\mathrm{eff}}\frac{\partial Q}{\partial y}\right) \approx \frac{\partial}{\partial y}\{\Delta\mathcal{A}[-\nabla\cdot(\mathbf{v}_\chi q)]\}. \tag{5.1}$$

A steady-state solution for $\partial Q/\partial y$ can be easily obtained by integration with respect to y, that is,

$$\frac{\partial Q}{\partial y} = \frac{\Delta\mathcal{A}[-\nabla\cdot(\mathbf{v}_\chi q)] + a}{K_{\mathrm{eff}}}, \tag{5.2}$$

where a is the integration constant.

To demonstrate the relative importance of the wave activity source versus K_{eff} in shaping the meridional profile of $\partial Q/\partial y$, we first prescribe the wave source forcing and K_{eff} diagnosed from the 240- and 30-km MPAS simulations and set $a = 0$ in (5.2). The resultant $\partial Q/\partial\phi_e$ profiles are shown as the black and red curves in Fig. 11a, respectively. The corresponding K_{eff} values are shown in Fig. 11b. Next, for each resolution, we flip the K_{eff} value from the other case and repeat the calculation (Fig. 11a, dashed curves). The 240-km solution (black solid line) has a sharper and more equatorward maximum of $\partial Q/\partial\phi_e$ than the 30-km solution (red solid line), consistent with the $\partial Q/\partial\phi_e$ distribution of the corresponding AGCM simulations. The exact peak values are somewhat higher than their AGCM counterparts, implying a negative integration constant a and hence positive $\overline{v_\psi' q'}$, if the peak values have to be adjusted to match those of the AGCM. Revealingly, the 240-km case with the 30-km K_{eff} (red dashed) resembles the 30-km case much more than the 240-km case, and also the 30-km case with the 240-km K_{eff} (black dashed) resembles the 240-km case much more than the 30-km case, indicating the predominant role of the effect diffusivity in shaping the meridional profiles of the vorticity and hence the upper-level jet.

A question naturally arises: What determines K_{eff} itself in the upper troposphere? Theoretical closure for effective diffusivity is an active research topic in the turbulence community, but existing theories are largely limited to homogeneous turbulence. Thus, the answer to the question will remain elusive until a sound parameterization of K_{eff} (likely in terms of $\partial Q/\partial y$, eddy phase speed, and zonal-mean wind) is developed to take into consideration the wave–turbulence duality of the eddy–mean flow interaction in the upper troposphere.

c. *Dynamical convergence via K_{eff}*

In both MPAS and CAM3, there is a strong suggestion of convergence on the eddy forcing for the jet stream as the resolution increases from the second highest to the highest resolutions (Figs. 7a and 8a). Given that the resolution sensitivity of the eddy vorticity forcing and the jet stream is closely linked to that of the effective diffusivity, the convergence of the former two should reflect the convergence of the latter. As explained in section 3, in the large Pe limit, K_{eff} of the type-II turbulence asymptotes to UL of the typical stirring scale, becoming independent of κ value itself. Thus, the Pe value itself can be approximated to be K_{eff}/κ.

To evaluate how close the simulations by the three AGCMs are to the large Pe limit, we plot in Fig. 12 the slope of $\log_{10}(K_{\mathrm{eff}}/\kappa)$ with respect to $\log_{10}(UL/\kappa)$ between every neighboring pair of resolutions against K_{eff}/κ (representative of Pe) at the higher resolution of the pair. All quantities here are computed based on

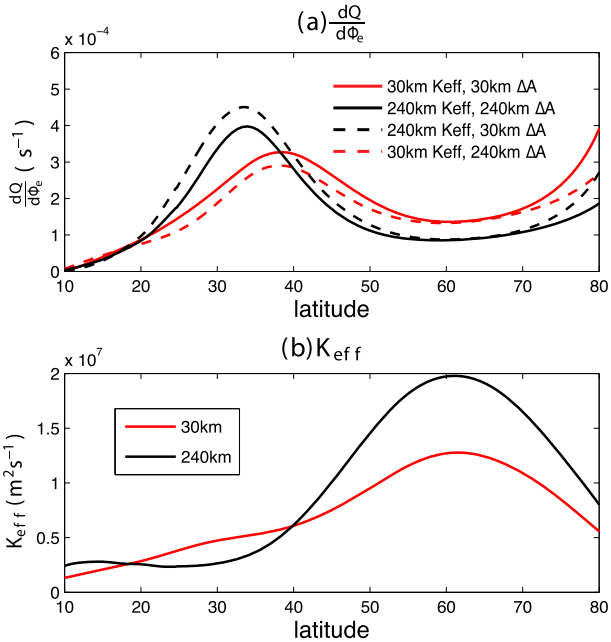

FIG. 11. (a) The solutions of $\partial Q/\partial \phi_e$ with K_{eff} and wave source prescribed alternately from the 30- and 240-km MPAS simulations; (b) K_{eff} from the MPAS simulations at 30- (red) and 240-km (black) resolutions.

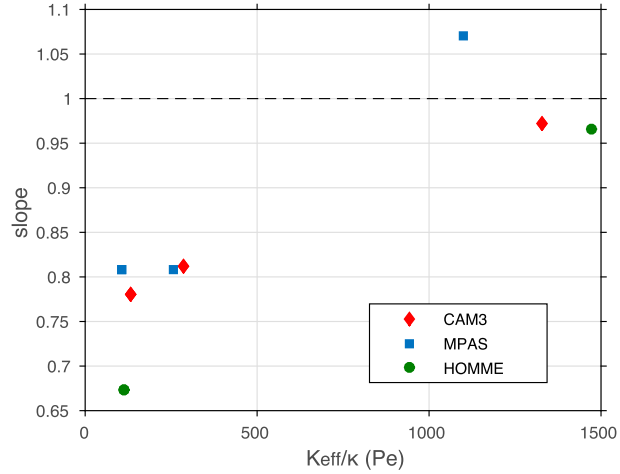

FIG. 12. The $\log_{10}(K_{\mathrm{eff}}/\kappa)$ vs $\log_{10}(UL/\kappa)$ slope as K_{eff}/κ increases with resolution in CAM3, MPAS, and HOMME (note that HOMME at 55 km is an outlier and hence not included in the slope estimation). All the quantities are based on the latitudinal average of K_{eff} and κ between 40° and 70°; UL is taken to be constant in the calculation. The unity slope indicates the independence of K_{eff} with regard to κ value.

K_{eff} and κ averaged over midlatitudes between 40° and 70°. To aid the interpretation of this figure, we reiterate that for the large Pe limit, the slope of $\log_{10}(K_{\mathrm{eff}}/\kappa)$ against $\log_{10}(UL/\kappa)$ should approach unity (indicating the independence of K_{eff} with regard to κ), while for the small Pe limit the slope should approach ½ [see Marshall et al. (2006) for the scaling argument for this limit].

Consistent with the scaling argument above, the slopes of $\log_{10}(K_{\mathrm{eff}}/\kappa)$ against $\log_{10}(UL/\kappa)$ estimated from all three AGCMs reside between 0.5 and 1 (approximately). For the slopes between the coarse-resolution pairs, they are clearly less than unity, indicating the dependence of K_{eff} with regard to κ. The slopes spanning between the second highest and the highest resolutions approach unity, with the corresponding Pe numbers being greater than 1000, which is about an order greater than the flow regime investigated in Marshall et al. (2006). For the slopes of the HOMME simulations, since the 55-km resolution is an outlier, the slope for the large Pe regime is estimated between the 28- and 110-km simulations. By omitting the case of 55 km, the conclusions from the analysis of MPAS and CAM3 seem to hold for HOMME as well.

As introduced in section 3, K_{eff}/κ equals the squared, normalized equivalent length $(L_{\mathrm{eq}}^2/L_0^2)$ and serves as a nondimensional metric for the effective resolution. One can see from Fig. 12 that at similar grid spacing among the highest resolutions of the three AGCMs, HOMME has higher effective resolution than CAM3, and CAM3 in turn has higher effective resolution than MPAS (consistent with the impression about the effective resolution obtained from the EKE spectrum; M. Taylor 2014, personal communication). Given the similar κ value for both 28-km HOMME and 30-km MPAS, K_{eff} itself is also larger in the former in the midlatitudes (cf. Figs. 9e and 7e). By inspecting Figs. 7e and 8e, one can see that K_{eff} peaks at approximately $10^7\,\mathrm{m^2\,s^{-1}}$ in the midlatitude when the resolution is about 0.5° and higher. The $10^7\,\mathrm{m^2\,s^{-1}}$ magnitude of K_{eff} coincides with what one might expect from the stirring by a synoptic scale with $U_s \sim 10\,\mathrm{m\,s^{-1}}$ and $L_s \sim 1000\,\mathrm{km}$. This likely convergence value remains to be verified with similar simulations using other aquaplanet AGCMs, as arguably the AGCMs explored here have a common lineage in physics, as well as with observations.

Redoing the calculation of K_{eff} and κ using half of the data gives results indistinguishable from Fig. 12, so it is unlikely that the estimate of the slopes in Fig. 12 is subject to sampling error.

6. Concluding remarks

A finite-amplitude wave activity budget equation is developed based on the vorticity equation for the upper

troposphere to diagnose the jet dynamics in three different aquaplanet AGCMs. In all three models, MPAS, CAM3, and HOMME, the eddy-driven westerlies show a consistent sign of convergence toward more poleward position with increasing resolution. In addition, the jet stream core intensity tends to decrease with resolution. Both behaviors can be rationalized by the asymptotic tendency of the effective diffusivity (K_{eff}) toward its limit value set by the unique property of the Batchelor turbulence that K_{eff} approaches the UL of the stirring scales (which is smaller than the energy containing scales) in the large Peclet number limit. Estimation from these aquaplanet AGCMs suggests a limit value of K_{eff} just shy of $10^7\,\mathrm{m}^2\,\mathrm{s}^{-1}$ for ∇^2 diffusion near its midlatitude peak. It remains to be seen whether this will be brought to bear in simulations with more realistic configuration and observations, wherein land–sea contrast and topography may inject wave activity and enstrophy at small scales. A possible rule of thumb gained from this exercise is that aquaplanet AGCMs with grid spacing around or finer than 50 km might be adequate for the modeled effective diffusivity to converge with consequential implications in the midlatitude jet stream and the associated variability.

A working hypothesis can be formed from the wave activity budget for the mechanism of the shift of the eddy-driven jet as the resolution increases. Compared to the coarse-resolution runs, the effective diffusivity in high resolution is smaller; more waves generated from baroclinic instability in the storm track can survive the dissipation and reach the free propagating level in the upper troposphere–lower stratosphere. This is confirmed by the higher wave activity level and the larger equatorward propagation of the wave activity in the higher-resolution simulations of the model. Those waves that cross the jet will finally break and get dissipated at the equatorward flank of the jet near their critical latitudes. Both the arrival of more wave activity and better resolution for the equivalent length in the subtropics lead to a greater effective diffusivity there, which works to mix the vorticity more extensively. As a result, the well-mixed region is expanded poleward and the peak of the vorticity gradient is broadened and weakened; the peaks of the eddy vorticity flux and the Lagrangian PV gradient are also pushed poleward, and so is the eddy-driven jet.

On the other hand, one should not attribute entirely the sensitivity of the jet location and intensity shown in Fig. 1 to the dynamics. The model physics and its interaction with the dynamics can be dependent on the resolution as well, bearing in mind, for example, the convection schemes in those models are not scale-aware (O'Brien et al. 2013). For similar grid-scale resolutions, MPAS and HOMME put the jet at a more poleward positions than CAM3, a possible reflection of the improvement of the convection scheme in the CAM4 physics. Nevertheless, in view of the robust jet sensitivity seen in these aquaplanet models, it is expected that the jet positions in the climate models participating in CMIP3 and CMIP5 are mostly equatorward biased given their coarse horizontal resolutions. Further, we conjecture that as the typical resolution of the atmospheric models for the next phase of the Coupled Model Intercomparison Project (i.e., CMIP6) is going to reach approximately 50 km, the equatorward bias in the SH jet position will be further reduced.

This study has been only focused on the grandest feature of the midlatitude circulation—the jet stream. The natural next step is to apply similar diagnostic framework to the extreme phenomena associated with anomalous wave activities such as blockings and atmospheric rivers. Compelling evidence abounds that changing characteristics of the jet stream can have profound impacts on the temporal and spatial distribution of these extremes (e.g., Ryoo et al. 2013; Anstey et al. 2013). To the extent that many of these extreme phenomena are governed by the enstrophy and energy cascades within the Batchelor regime, the concept of effective diffusivity and the related diagnostics can be equally useful for quantifying the temporal characteristics and predictability of these extremes.

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APPENDIX A

Proof of $\Delta\mathcal{A}(\mathbf{v}_\psi \cdot \nabla q) = \overline{v'_\psi q'}$

Applying the area-coordinate transformation operator (3.2) to the advection term,

$$\Delta\mathcal{A}(\mathbf{v}_\psi\cdot\nabla q)=\frac{1}{2\pi a\cos\phi_e}\left(\iint_{q>Q}\mathbf{v}_\psi\cdot\nabla q\,dS-\iint_{\phi>\phi_e}\mathbf{v}_\psi\cdot\nabla q\,dS\right). \tag{A1}$$

The first term in the parentheses is

$$\begin{aligned}\iint_{q>Q}\mathbf{v}_\psi\cdot\nabla q\,dS&=\iint_{q>Q}\nabla\cdot(\mathbf{v}_\psi q)\,dS\\&=\oint_Q q\mathbf{v}_\psi\cdot\mathbf{n}\,dl\\&=Q\oint_Q\mathbf{v}_\psi\cdot\mathbf{n}\,dl=0,\end{aligned} \tag{A2}$$

where the subscript Q indicates that the line integral is along a constant Q contour; thus, Q can be taken out of the integral. In deriving (A2) divergence theorem and the nondivergence of $\mathbf{v}_\psi$ have been used. It is straightforward to show that the second term of (A1) is $\overline{v'_\psi q'}$ as follows:

$$\begin{aligned}&\frac{1}{2\pi a\cos\phi_e}\iint_{\phi>\phi_e}\mathbf{v}_\psi\cdot\nabla q\,dS\\&=\frac{1}{2\pi a\cos\phi_e}\iint_{\phi>\phi_e}\nabla\cdot(\mathbf{v}_\psi q)\,dS\\&=\frac{1}{2\pi a\cos\phi_e}\oint_{\phi_e}\mathbf{v}_\psi q\,dl=\overline{v_\psi q}=\overline{v'_\psi q'}.\end{aligned} \tag{A3}$$

The last step is derived from the fact that the zonal mean of the rotational component of the wind $\overline{v_\psi}=0$.

From (A2) one can infer

$$\iint_{q>Q}\frac{\partial Q}{\partial t}\,dS=0 \tag{A4}$$

for a conservative tracer advected by two-dimensional rotational flow, meaning that no vorticity substance enters and leaves the area bounded by Q.

APPENDIX B

Derivation of Effective Eddy Diffusivity K^e_{eff}

The concept of effective eddy diffusivity arises from a different area-coordinate operator from the original formalism of effective diffusivity of Nakamura (1996). Interested viewers are referred to Chen and Plumb (2014) for the original definition and physical interpretation of effective eddy diffusivity.

Applying the area transformation operator (3.2) to the diffusion term,

$$\begin{aligned}\Delta\mathcal{A}(\kappa\nabla^2 q)&=\frac{1}{2\pi a\cos\phi_e}\left[\iint_{q>Q}(\kappa\nabla^2 q)\,dS-\iint_{\phi>\phi_e}(\kappa\nabla^2 q)\,dS\right]\\&=\frac{1}{L_0}\oint_Q\kappa\nabla q\cdot\mathbf{n}\,dl-\frac{1}{L_0}\oint_{\phi_e}\kappa\nabla q\cdot\mathbf{n}\,dl\\&=\frac{1}{L_0}\oint_Q\kappa\nabla q\cdot\frac{\nabla q}{|\nabla q|}dl-\frac{1}{L_0}\oint_{\phi_e}\kappa\nabla q\cdot\frac{\nabla q}{|\nabla q|}dl\\&=-\frac{1}{L_0}\frac{\partial}{\partial Q}\iint_{q>Q}\kappa\nabla q\cdot\nabla q\,dS+\frac{1}{L_0}\oint_{\phi_e}\kappa\frac{\partial q}{\partial y}dx\\&=-\frac{1}{L_0}\frac{\partial y}{\partial Q}\oint_Q\kappa|\nabla q|^2\,dl+\frac{1}{L_0}\oint_{\phi_e}\kappa\frac{\partial q}{\partial y}dx\\&=-\frac{\langle\kappa|\nabla q|^2\rangle_Q}{(\partial Q/\partial y)^2}\frac{\partial Q}{\partial y}+\kappa\frac{\partial\overline{q}/\partial y}{\partial Q/\partial y}\frac{\partial Q}{\partial y}\\&=-\left(K_{\mathrm{eff}}-\kappa\frac{\partial\overline{q}/\partial y}{\partial Q/\partial y}\right)\frac{\partial Q}{\partial y},\end{aligned} \tag{B1}$$

where the subscripted angle bracket denotes the average along the Q contour. From the third to the fourth expresssion, the Gauss divergence theorem is used (the minus sign is due to the fact that $\nabla q/|\nabla q|$ points to the inside of the circuit); from the fourth to the fifth expression, the chain rule

$$\frac{\partial}{\partial Q}=\frac{\partial}{\partial y}\frac{\partial y}{\partial Q}$$

is used. Note that K_{eff} is the same as the original effective diffusivity for the regular diffusion ($m=1$) in Nakamura (1996) and Nakamura and Zhu (2010).

As in Chen and Plumb (2014), we define the effective eddy diffusivity as

$$K^e_{\text{eff}} \equiv K_{\text{eff}} - \kappa \frac{\partial \overline{q}/\partial y}{\partial Q/\partial y}. \tag{B2}$$

Defined as such, K^e_{eff} represents the diffusivity due to the net of the diffusive wave activity flux through the bound demarcated by the tracer contour Q (i.e., $-K_{\text{eff}}\partial Q/\partial y$) minus that through the latitude ϕ_e (i.e., $\kappa\partial\overline{q}/\partial y$).

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